DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-ITCS226L
COURSE TITLE	**:**	Information Management Laboratory
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 unit
LEARNING MODALIT	**:**	Fully Onsite
PRE-REQUISITES	**:**	S-ITCS123/L
CO-REQUISITES	**:**	S-ITCS226LA
PROFESSOR	**:**	Arnold B. Galve, DITc
CONSULTATION HOURS	**:**	Monday: 18:00 – 1930; Online
CONTACT INFORMATION:		abgalve@dlsud.edu.ph

VISION
A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION
To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:
This course focuses on data organization and architecture as well as managing the database. Topics include the basic functions of a data administrator, including database planning, analysis, design, implementation, maintenance and protection.

COURSE LEARNING OUTCOMES:
By the end of the semester, the students can:

CLO1. analyze business domains and identify appropriate database architecture and data model;
CLO2. select suitable schema in designing and developing database solution architecture for an organization;
CLO3. simulate the functions of database administrators throughout the information lifecycle;
CLO4. administer data capture, management, storage and retrieval method;
CLO5. implement appropriate database solution for an organization/company/department/unit; and
CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN: (Midterm Period)

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Analyze business domains and identify appropriate database architecture and data model	TLO1. Identify and visualize different database characteristics TLO2.Recognize and understand the basic concepts and constructs of a data model.
CLO2. Select suitable schema in designing and developing database solution architecture for an organization.	TLO3. Understand strategies and parameters for different design phase TLO4. Describe a useful tool for mapping data requirements to business operating locations TLO5. Select phases of database design process
CLO3. Simulate the functions of database administrators throughout the information lifecycle.	TLO6. Analyze and implement appropriate data model. TLO7. Correlate logical database to physical database TLO8. Apply and manipulate SQL Commands
CLO6. Demonstrate 21st century skills in all learning activities.	

Midterm Period:

Module no. and Learning Outcomes	Week / Date(s)	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Allotted
Module 0 CLO 1 TLO 1	Week1 Jan 20 to Jan 25	▪ Gospel, Reflection and Prayer ▪ Discussion of Course Syllabus **Module 0 – Course Orientation**	Schoolbook	3.0
Module 1 CLO1 TLO1 TLO2	Week2 Jan 27 to Feb 1	▪ Gospel, reflection and Prayer ▪ Class discussion on Introduction to Information Management ▪ *Formative Assessment:* Familiarization and Warm-up exercise using SmartDraw **Module 1 – Introduction to Information Management**	Schoolbook SmartDraw	3.0
Module 2 CLO1 CLO2 TLO1 TLO2	Week3 Feb 3 to Feb 8	▪ Gospel, Reflection and Prayer ▪ Explain the concept of databases ▪ Discuss the process of data management ▪ Explain various types of DBMS	Schoolbook	3.0

		▪ *Enabling Assessment:* Data Model design using SmartDraw **Module 2 – Databases and Information Management**		
Module 3 CLO3 TLO3 TLO4	Week4 Feb 10 to Feb 15	▪ Gospel, reflection and Prayer ▪ Explain how database modelling is done in databases ▪ Discuss the importance of database modelling ▪ *Enabling Assessment:* Entity Relationship Model of Sales and Inventory System **Module 3 – Database Modelling**	Schoolbook SmartDraw MS Access	3.0
Module 4 CLO3 TLO3 TLO4	Week5 Feb 17 to Feb 22	▪ Gospel, reflection and Prayer ▪ Explain the concept of normalization ▪ Demonstrate how normalization is done in databases ▪ *Enabling Assessment:* Entity Relationship Model of a Library System **Module 4 – Database Normalization**	Schoolbook SmartDraw MS Access	3.0
Module 5 CLO2 CLO3 TLO8 TLO9	Week6 Feb 24 to Mar 1	▪ Gospel, reflection and Prayer ▪ Discuss relational database concepts. ▪ Demonstrate how to create relations and mapping from ERD to relations. ▪ Show how to set integrity rules in databases ▪ Explain different integrity rules. ▪ *Enabling Assessment:* ERD of an Enrolment System Database **Module 5 – Relational Database**	Schoolbook SmartDraw MS Access	3.0
Module 6 CLO3 TLO6 TLO7	Week7 Mar 3 to Mar 8	▪ Gospel, reflection and Prayer	Schoolbook SmartDraw MS Access	3.0

		▪ Discuss the importance and impacts of securing the database. ▪ Explain various types of enforcing security on databases. ▪ *Enabling Assessment:* Enforcing Integrity Rules in Enrolment Database **Module 6 – Database Security**		
Midterm Exam Week	Week8 Mar 10 to Mar 15	**MIDTERM SUMMATIVE ASSESSMENT**		3.0
			TOTAL	**24**

LEARNING PLAN: (Final Period)

Course Learning Outcomes	**Topic Learning Outcomes**
CLO3. Simulate the functions of database administrators throughout the information lifecycle	TLO9. Distinguish different user roles in database. TLO10. Determine the applicable security measure in different applications.
CLO4. Administer data capture, management, storage and retrieval method.	TLO8. Apply and manipulate SQL Commands
CLO5. Implement appropriate database solution for an organization / company / department/unit.	TLO6. Analyze and implement appropriate data model. TLO7. Correlate logical database to physical database TOL8. Apply and manipulate SQL commands
CLO6. Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Final Period:

Module no. and Learning Outcomes	Week / Date(s)	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Allotted Hours
Module 7 CLO2 CLO3 TLO8 TLO9	Week10 Mar 24 to Mar 29	**Sustainability Week** Completion / Consultation Week **Module 7 – Information Retrieval**	Schoolbook MS Teams	✓	3.0
Module 7 CLO2 CLO3 TLO8 TLO9	Week11 Mar 31 to Apr 5	▪ Gospel, reflection and Prayer ▪ Explain the different queries that select data, sort and filter data and use aggregate data. ▪ *Enabling Assessment:* Application of information retrieval commands. **Module 7 – Information Retrieval continuation**	Schoolbook	✓	3.0
Module 8 CLO4 TLO8	Week12 Apr 7 to Apr 5	▪ Gospel, reflection and Prayer ▪ Explain and demonstrate how backup and recovery is done in databases. ▪ *Enabling Assessment:* Research matrix on database backup and recovery tools. **Module 8 – Database Backup and Recovery**	Schoolbook	✓	3.0
	Week13 Apr 14 to Apr 19	▪ Gospel, reflection and Prayer ▪ Continuation of discussion for module 8. ▪ *Enabling Assessment:* How to backup and recover a damaged/corrupted database.	Schoolbook	✓	3.0
Module 9 CLO3 CLO4 TLO6	Week14 Apr 21 to Apr 26	▪ Gospel, reflection and Prayer ▪ Explain and demonstrate how database maintenance is done in RDBMS ▪ *Enabling Assessment:* Maintaining a database of an organization	Schoolbook	✓	3.0

		Module 9 – Database Maintenance			
Module 10 CLO3 CLO4 TLO7 TLO8	Week15 Apr 28 to May 3	▪ Gospel, reflection and Prayer ▪ Discussion about introduction to SQL. ▪ Demonstrate various SQL commands. ▪ *Enabling Assessment:* Machine problem on structured query commands **Module 10 – Introduction to SQL**	Schoolbook XAMPP PowerPoint	✓	3.0
	Week16 May 5 to May 10	▪ Gospel, reflection and Prayer ▪ Continuation of discussion for module 10. ▪ *Enabling Assessment:* Machine problem on advanced structured query commands	Schoolbook XAMPP PowerPoint	✓	3.0
Final Exam Week	Week17 May 12 to May 17	**FINAL SUMMATIVE ASSESSMENT**		✓	3.0
			TOTAL	**8**	**24**

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments (Activities, Quizzes, Presentations, etc.)	40%	Enabling Assessments (Activities, Quizzes, Presentations, etc.)	40%
Summative Assessment (Midterm Examination)	60%	Summative Assessment (Final Examination)	60%
	100%		**100%**

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly

prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Aklatang Emilio Aguinaldo

Hoffer, J, Ramesh, V., Topi, H. 2020. modern database management. 11th edition. Pearson Education South Asia PTE. LTD

Morrow, Jordan. 2022. be data driven: how organizations can harness the power of data.

Database Principles: fundamentals of design, implementation and management. 10th edition.

Online

Tutorial: Get Started with Tableau Desktop. helptableaucom. https://help.tableau.com/current/guides/get-started-tutorial/en-us/get-started-tutorial-home.htm.

SQL with Microsoft Access 2016 Lesson 5 - AND, OR, LIKE. wwwyoutubecom. [accessed 2024 Jan 19]. https://www.youtube.com/watch?v=xqSTroBqyeQ&list=PLzr5fRV1AGV8dEGsRkghCpXNGEFI1PFaA&index=5.

Tutorial: Get Started with Tableau Desktop. helptableaucom. https://help.tableau.com/current/guides/get-started-tutorial/en-us/get-started-tutorial-home.htm.

SQL with Microsoft Access 2016 Lesson 5 - AND, OR, LIKE. wwwyoutubecom. [accessed 2024 Jan 19]. https://www.youtube.com/watch?v=xqSTroBqyeQ.

SQL with Microsoft Access 2016 Lesson 10 - Group By, Max, Min, Average. wwwyoutubecom. [accessed 2024 Jan 19]. https://www.youtube.com/watch?v=Ea5PyitZndE&list=PLzr5fRV1AGV8dEGsRkghCpXNGEFI1PFaA&index=10.

EBSCOHOST (Journals)

Abadi S, Huda M, Hehsan A, Marzuki Mohamad A, Basiron B, Suhaila Ihwani S, Azmi Jasmi K, Safar J, Kilani Mohamed A, Hassan Wan Embong W, et al. 2018. Design of online transaction model on traditional industry in order to increase turnover and benefits. International Journal of Engineering & Technology. 7(2.27):231. doi:https://doi.org/10.14419/ijet.v7i2.27.11982.

Figueiredo MSN, Pereira AM. 2017. Managing Knowledge – The Importance of Databases in the Scientific Production. Procedia Manufacturing. 12:166–173. doi:https://doi.org/10.1016/j.promfg.2017.08.021.

Li J, Hui B, Qu G, Yang J, Li B, Li B, Wang B, Qin B, Geng R, Huo N, et al. Can LLM Already Serve as A Database Interface? A BIg Bench for Large-Scale Database Grounded Text-to-SQLs. [accessed 2024 Jan 19]. https://proceedings.neurips.cc/paper_files/paper/2023/file/83fc8fab1710363050bbd1d4b8cc0021-Paper-Datasets_and_Benchmarks.pdf.

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2nd Semester 2024 – 2025

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